

Architecting for Action: Data Lineage and Analysis via the T-DAR Framework

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Abstract

Data lineage is a critical aspect of modern data management, providing insights into the flow and transformation of data across complex ecosystems. This paper introduces the T-DAR framework, a novel approach to analyzing data lineage that emphasizes the relationship between targeted observations, data assets, and actionable results. By leveraging the T-DAR framework, organizations can enhance their understanding of data processes, improve decision-making, and ensure compliance with regulatory requirements.

1 Introduction

In the era of Big Data, organizations have adopted a data-driven approach to decision-making. However, the complexity of data ecosystems often leads to challenges in understanding the flow and transformation of data. This paper introduces the **T-DAR framework**, which stands for Targeted Data-Actions-Results, as a methodology for analyzing data lineage and ensuring actionable insights. The T-DAR framework provides a structured approach to capturing and analyzing the relationships between data assets, the actions taken on them, and the resulting outcomes. By focusing on targeted observations, organizations can gain a deeper understanding of their data processes and make informed decisions based on actionable insights.

2 T-DAR Framework Overview

The T-DAR framework is inspired by the Data-Action-Result (DAR) model of most healthcare information systems.

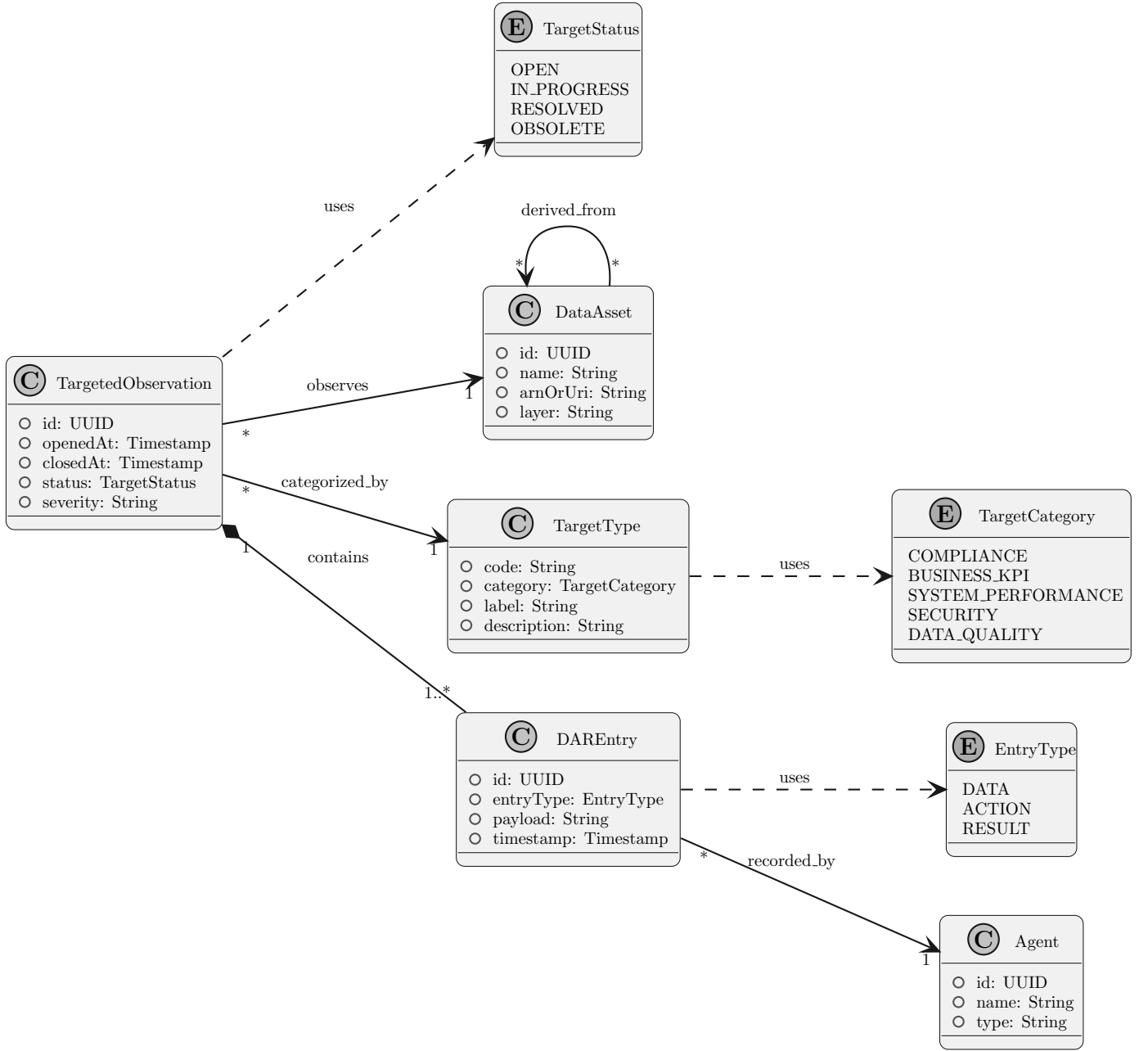


Figure 1: T-DAR Architecture Overview

As illustrated in Figure 1, the architecture of the T-DAR framework is centered around a *Targeted Observation*, which is directly linked to a specific *Data Asset*

within the ecosystem. Rather than treating data lineage as a static representation of data streams, the T-DAR framework takes advantage of the dynamic nature of data environments, allowing for real-time analysis and decision-making. It provides a way to analyse not only results, but also the medium through which the results are obtained, and the actions that lead to those results.

To operationalize this dynamic analysis, every *Targeted Observation* acts as a container for a chronological sequence of *DAREntry* records. These entries are categorized into three distinct phases: *Data*, *Action*, and *Result*, which together form the foundation of the framework:

- **Data (D) – The Assessment:** A Data entry represents a factual snapshot, an audit trail, or a metric collected from the *Data Asset*.
- **Action (A) – The Intervention:** An Action entry represents a specific action taken in response to the data or the results of the data.
- **Result (R) – The Outcome:** A Result entry represents the outcome or consequence of the preceding action taken.

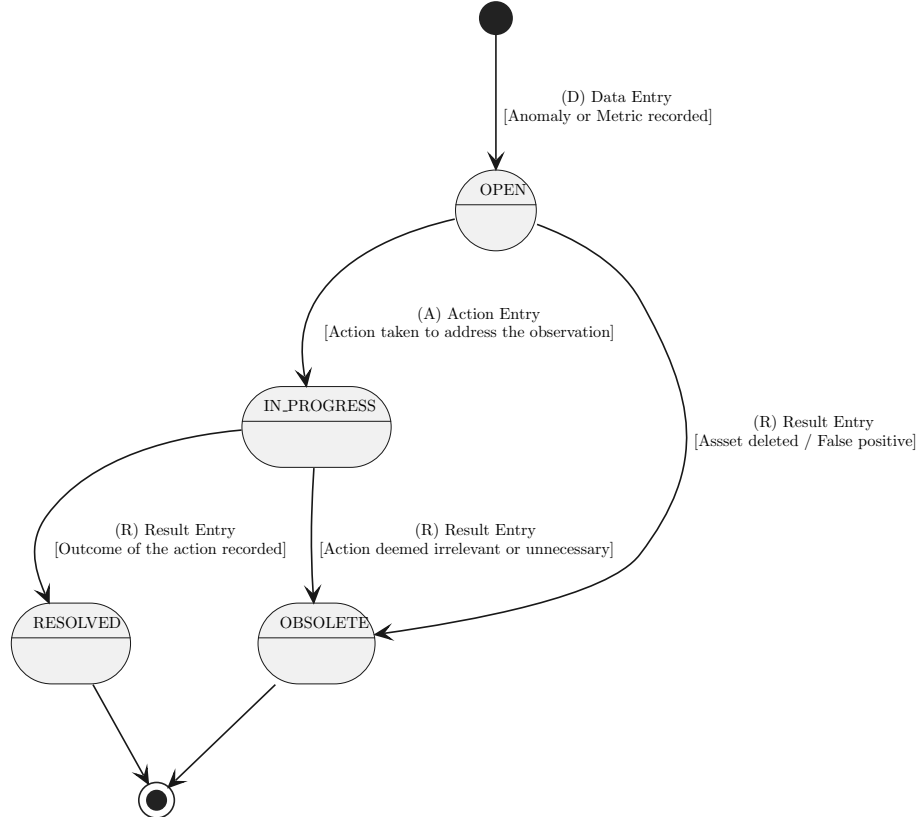


Figure 2: T-DAR State Diagram

As shown in Figure 2, these DAR entries act as triggers that drive the lifecycle of a *Targeted Observation*, transitioning it from an initial open state to a final resolution.

Furthermore, since data assets are inherently interconnected through a **derived** **_from** lineage relationship (as modeled in the framework architecture), the T-DAR framework allows organizations to track how an *Action* applied to an upstream asset propagates its *Results* to downstream dependencies. This elevates traditional data lineage from a simple dependency graph to an actionable governance tool.

3 Case Study: Applying T-DAR to a Medallion Architecture

To demonstrate the practical value of the T-DAR framework, we consider a typical data engineering pipeline organized in a medallion architecture, which consists of three layers: *Bronze* (Raw), *Silver* (Cleaned), and *Gold* (Aggregated).

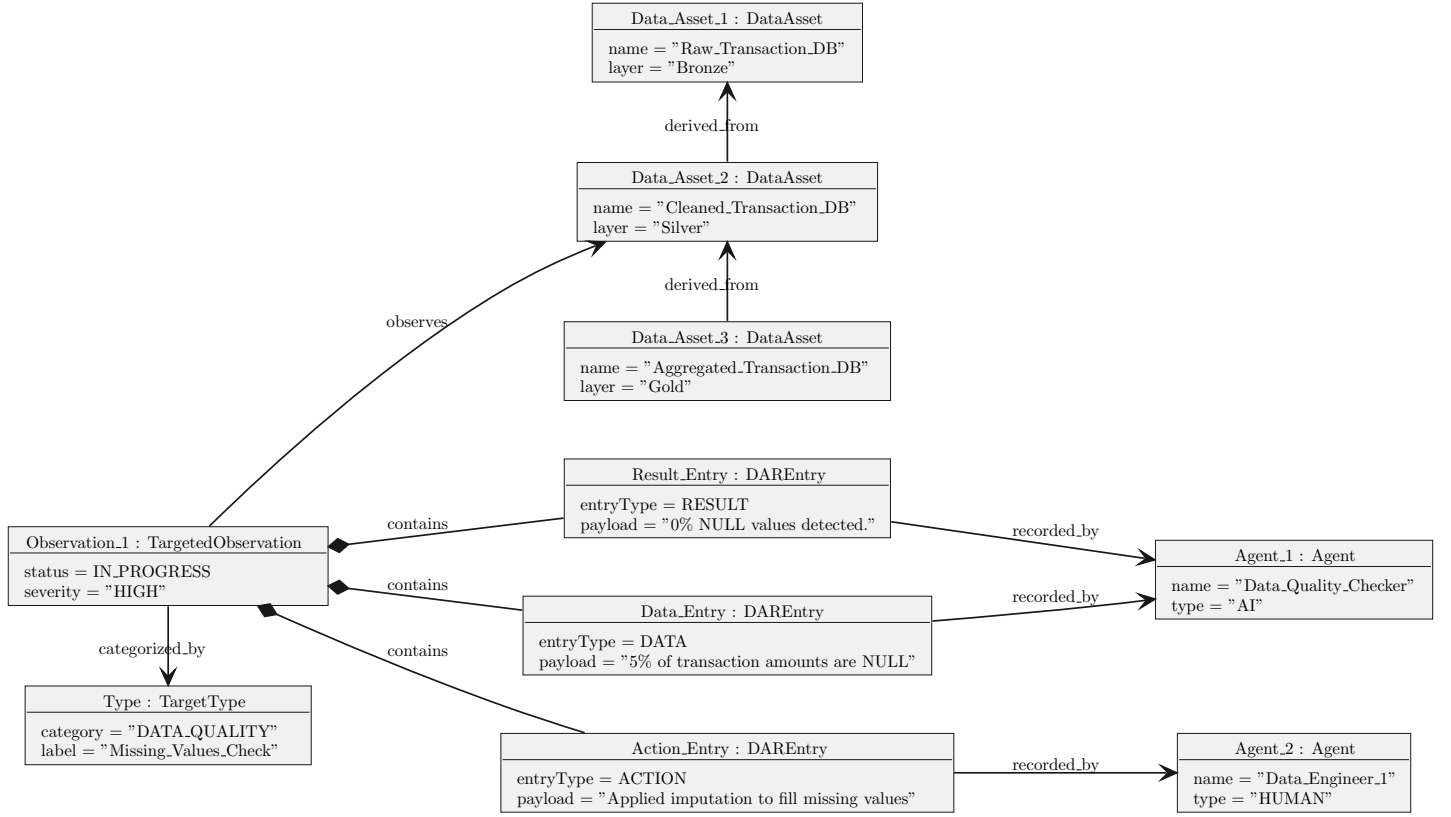


Figure 3: T-DAR Applied to a Medallion Architecture

As depicted in Figure 3, the framework provides end-to-end traceability when a data quality issue occurs. In this scenario, an AI agent (*Agent_1*) detects that 5% of the transaction amounts are missing in the *Silver* layer. This event triggers the creation of a *Targeted Observation* along with a *Data Entry* that records the issue.

Because the system explicitly records the **derived_from** lineage, data stewards immediately know that the downstream *Gold* layer is also affected by this anomaly. Subsequently, a human data engineer (*Agent_2*) is assigned to address the issue. The engineer applies an imputation strategy to fill in the missing values, which is recorded as an *Action Entry*. Finally, a *Result Entry* captures the outcome of this intervention, confirming that the data quality issue has been resolved.

3.1 Implementation Example: T-DAR API Payload

To illustrate how the framework integrates with modern data stacks, the following listing provides an example of the JSON payload generated by the automated AI agent (*Agent_1*) to initialize the observation on the *Silver* layer. This standard structure allows data engineering tools to seamlessly push governance metadata into the T-DAR system.

Listing 1: JSON payload for creating a Data Entry

```
{
  "observation_id": "obs-1",
  "target_type": "DATA_QUALITY",
  "data_asset_id": "asset-silver-db",
  "status": "OPEN",
  "dar_entry": {
    "entry_type": "DATA",
    "agent_id": "ai-checker-bot",
    "timestamp": "2026-09-22T10:00:00+02:00",
    "payload": {
      "metric": "null_values_percentage",
      "value": 0.05,
      "message": "5% of transaction amounts are NULL"
    }
  }
}
```